

6.2 Creating distributions from data

Watch: Visualising variability (part 1) [15.35 mins]

Watch the following video, which explains various terms and describes ways in which the frequency distribution of a categorical variable can be visualised. Benford law is also explained.

This is useful as it provides a fundamental understanding of visualising categorical data distribution.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138520?wrap=1>

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Download the following slides for note taking: [ICT702 Visualising variability \(part 1\)](https://learn.usc.edu.au/courses/35242/files/3138665?wrap=1)
<https://learn.usc.edu.au/courses/35242/files/3138665?wrap=1>

There are several aspects to consider when creating visualisations of data distributions. The following defines key terms before moving on to a consideration of the different types of data.

Variation

Almost every challenge that is faced in business is impacted by knowing and understanding the potential values that relevant variables may have on the outcome of that, which is of interest in the

decision-making process. Values are impacted criteria. Values of variables can vary. For example, aspects such as when they are measured (time), who is being measured (e.g. customer range) and what is being measured (product variation) may make a difference. This is called *variation*. Variation introduces uncertainty, which is not controllable. The term *random variable* refers to a quantity whose value is not able to be ascertained.

Descriptive analytics

Data that has been collected consists of values that are known. The data is analysed and visualised to gain a better understanding of the variation and the impact that may result. In general, we are not able to collect data from the whole population of interest. Therefore, the technique of sampling is employed to enable data to be collected from a subset of the population. If the subset is representative of the population, then generalisations of the whole population can be inferred. This is called *descriptive analytics*.

Frequency distribution

After data is collected, it is often summarised to illustrate how often specific types of data values appear. This is called *frequency distribution*. Frequency distribution visualisations can be developed for either *categorical data* or *quantitative data*.

Categorical data

Categorical data consists of labels or names; it is not possible to mathematically manipulate this type of data. When analysing categorical data, the data is typically separated into non-overlapping groups, known as bins or classes. The bins are then used to create a frequency distribution. Once the frequency distribution has been visualised (e.g. as a bar or column chart) it can be ordered (e.g. from largest to smallest). However, ordering the data is not appropriate if the bins of data have ordinal relationships. Where there are ordinal relationships between the data, those relationships should be maintained in the visualisation.

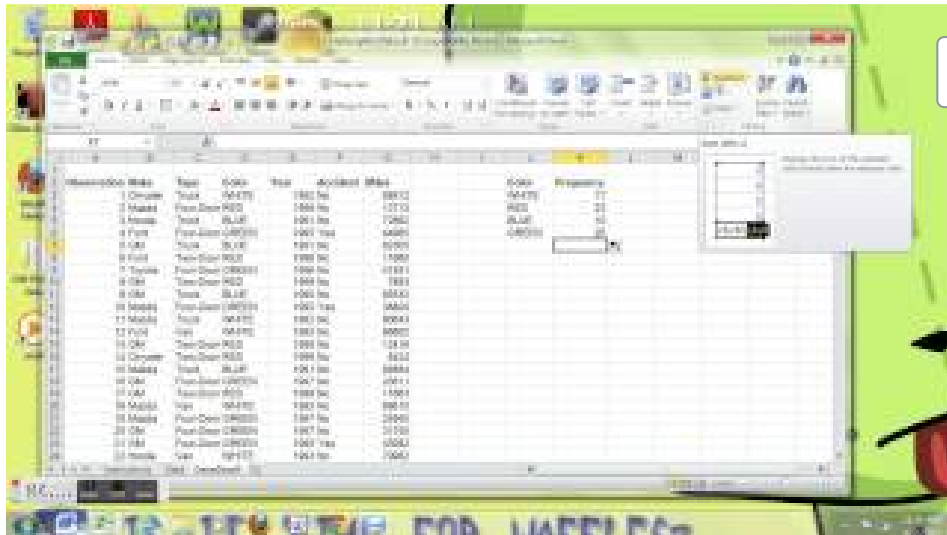
Frequency distributions for categorical data

□ Watch: Frequency distributions: Categorical data [2.56 mins]

Watch the following video to learn about frequency distributions. This video is useful because it demonstrates how to use MS Excel to calculate the frequency, relative frequency and percentage frequency of a categorical or qualitative variable.

As you watch consider:

- How does this help you to create useful visualisations of categorical data?
- Why is this important?



Source: © Alex Vernon (2013).

Relative frequency refers to the fraction or proportion of items that belong to a bin/class. Relative frequency is often expressed as a percentage (i.e. relative frequency multiplied by 100).

Percent frequency distribution can be used to estimate the relative likelihood of different values of a random variable. This means that the probability distribution, which characterises the variability of a random variable, can be estimated.

Visualising distributions of quantitative data

□ Watch: Visualising variability (part 2) [15.42 mins]

The video below explains that frequency distributions can also be created for quantitative data and describes how this is achieved. Various types of charts and techniques that are useful for visualising and comparing quantitative distributions are introduced. Further, the shape of a distribution and skewness are also discussed.

This is useful as it builds understanding of frequency distribution and explains how visualisations of quantitative distributions can be created.



Transcript

<https://learn.usc.edu.au/courses/35242/files/3138521?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Visualising variability \(part 2\)](https://learn.usc.edu.au/courses/35242/files/3138574?wrap=1)
(<https://learn.usc.edu.au/courses/35242/files/3138574?wrap=1>)

Quantitative data

Quantitative data consists of numerical values, and it is possible to mathematically manipulate this type of data. Quantitative data can also be grouped into bins to create distributions of frequency. However, you must take care that the bins are non-overlapping. For quantitative data, the bins are based on ranges of values, such as age ranges; therefore, you must make sure that adjacent bins do not include any of the same values.

The notation of the bins is important – bin ranges are notated as follows (17, 25]. The round bracket signifies that the value is not included in the bin and the square bracket signifies that the value is included in the bin.

In order to create a frequency distribution for quantitative data, three elements have to be defined. These are the:

1. number of non-overlapping bins
2. width (numerical range) of each bin
3. range spanned by the set of bins.

The choice of the number of bins and the bin width can impact how a distribution displays in a visualisation.

Number of bins

This is formed from specifying the ranges of each group of data; a total of five to 20 bins is ideal. Using too few bins results in too much aggregation of data and the variation in the data is blurred, not captured accurately and only produces high-level patterns. Alternatively, using too many bins results in generalisable patterns not being captured. The “best” number of bins is often determined by the subject matter and analytical goal.

Width of the bins

There should be a uniform width across the range spanned by the bins. The number of bins and the width of the bins are related; less bins generally means greater bin width, etc. Therefore, these elements may be determined by trial and error, with a number of variations tested.

Range spanned by the bins

It is important to make sure that the set of bins includes the whole range of the data. Equally, as stated above, it is important to make sure that each value is only included in one bin.

Techniques for visualising distributions

Distributions are often visualised as *histograms*. A histogram is a column chart without spaces between the columns. Each column represents a bin, and the height of each column represents the frequency of the values in the relevant bin. Histograms provide information about the shape/form of a distribution.

› **Skewness**

› **Frequency polygons**

› **Trellis displays**

› **Strip charts**

› **Occlusion**

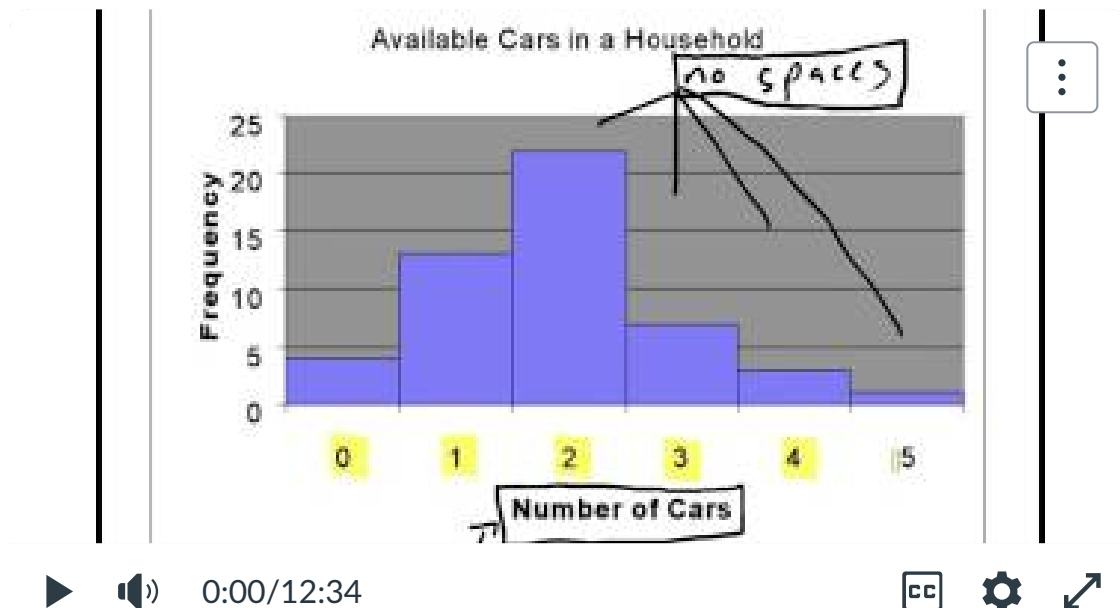
› **Jittering**

□ Watch: Displaying a quantitative variable [12.33 mins]

Watch the video below on histograms. It explains what histograms are and how they can be used. This is useful, because histograms are frequently used to create visualisations of quantitative data.

As you watch consider:

- how this enables your ability to create useful visualisations of quantitative data
- why understanding skewness is important.



Source: © Sulock Math (2017).

□ Activity: How global housing prices have changed since 2010

Read the following article on how global housing prices have changed in the last decade.

As you read, consider and make notes on:

- Which of the data visualisations was easier for you to understand?
- What did you like/not like about the first visualisation?
- How would you approach creating a visualisation for the data?
- What techniques would you use, and why?

Explain your reasoning and bring your notes to the workshop.

Soltani, E. (2023, April 2). [How global housing prices have changed since 2010.](https://www.visualcapitalist.com/cp/mapped-global-housing-prices-since-2010/)
[\(https://www.visualcapitalist.com/cp/mapped-global-housing-prices-since-2010/\)](https://www.visualcapitalist.com/cp/mapped-global-housing-prices-since-2010/)
Visual Capitalist.